

Cauchy Euler & System(Edit)

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Augustin Louis Cauchy

Quick Info

Born

21 August 1789

[Paris, France](#)

Died

23 May 1857

Sceaux (near Paris), France

Summary

Augustin-Louis Cauchy pioneered the study of analysis, both real and complex, and the theory of permutation groups. He also researched in convergence and divergence of infinite series, differential equations, determinants, probability and mathematical physics.

- **Rigor in Calculus:** He provided the first clear and precise definitions of **limits, continuity, and convergence**, moving calculus from intuitive guesswork to a rigorous science.
- **Complex Analysis:** Almost single-handedly founded this field. He developed the **Cauchy-Riemann equations** (necessary for functions to be differentiable) and **Cauchy's Integral Theorem**, which is essential for modern physics and engineering.
- **Theory of Elasticity:** Founded the mathematical theory of elasticity. He introduced the **Cauchy stress tensor**, a 3x3 matrix used to describe the internal state of stress at a point in a material.
- **Continuum Mechanics:** He formulated the fundamental equations of motion for continuous media,

treating matter as a continuum rather than discrete points.

- **Group Theory:** He was a pioneer in the study of **permutation groups**, laying the early groundwork for abstract algebra.
- **Fermat's Polygonal Number Theorem:** In 1812, he was the first to prove this difficult conjecture in number theory, which states every positive integer is a sum of orthogonal numbers.
- **Cauchy's Dispersion Equation:** In optics, he created an empirical formula that describes how the **refractive index** of a transparent material depends on the wavelength of light.
- **Problem of Apollonius:** He discovered a simple, elegant solution for describing a circle that touches three given circles.
- **Determinants & Matrices:** He invented what is now called the **Jacobian** (in two and three dimensions) and formulated various theorems on the inverse of a matrix and subdeterminants.
- **Calculus of Residues:** He invented this specific technique for evaluating complex integrals, which is now a standard tool in complex analysis.
- **Wave Propagation:** Won a Grand Prix in 1816 for his memoir on waves at the surface of a liquid, which became a classic in hydrodynamics.



Leonhard Euler is considered the most prolific mathematician in history, with his collected works spanning over **80 volumes**. His contributions defined the language of modern mathematics and fundamentally advanced physics, astronomy, and engineering.

Mathematical Notation & Functions

Euler practically invented the "language" of mathematics used today.

- **Functional Notation:** Introduced $f(x)$ to denote a function applied to an argument.
- **Fundamental Constants:** Defined or popularized e (base of natural logarithms), i (imaginary unit), and π (circle constant).
- **Other Standard Symbols:** Introduced Σ for summations, Δ for finite differences, and used a, b, c for triangle sides with A, B, C for angles.
- **Trigonometry:** Redefined sine, cosine, and tangent as functions rather than just lengths in a circle. [Wikipedia +4](#)

- **Complex Analysis:** Discovered **Euler's Formula** ($e^{i\varphi} = \cos\varphi + i\sin\varphi$) and **Euler's Identity** ($e^{i\pi} + 1 = 0$), linking algebra, trigonometry, and complex numbers.
- **Calculus of Variations:** Invented this field to find optimal solutions (e.g., the strongest shape for a beam) and derived the **Euler–Lagrange equation**.
- **Infinite Series:** Solved the famous **Basel Problem** (sum of $1/n^2 = \pi^2/6$) and pioneered the use of power series.
- **Special Functions:** Elaborated on the **Gamma function** (generalizing factorials) and the **Beta function**.
- **Differential Equations:** Developed **Euler's Method** for numerical integration and advanced the theory of linear differential equations. Wikipedia +6
- **Prime Numbers:** Proved the infinitude of primes using the divergence of the harmonic series and linked primes to the **Riemann zeta function** via the Euler product formula.
- **Totient Function:** Invented $\phi(n)$ and used it to generalize Fermat's Little Theorem into **Euler's Theorem**.
- **Theorems & Identities:** Proved Fermat's Last Theorem for $n = 3$, developed the theory of **perfect numbers**, and formulated the **law of quadratic reciprocity**. Wikipedia +2

Graph Theory & Topology

- **Founding Graph Theory:** Solved the **Seven Bridges of Königsberg** problem by focusing on nodes and edges rather than physical distance.
- **Euler Characteristic:** Discovered the formula $V - E + F = 2$ for convex polyhedra, which became a cornerstone of **topology**. Wikipedia +2

Physics & Applied Sciences

- **Fluid Dynamics:** Formulated the **Euler equations** for the motion of inviscid fluids.
- **Rigid Body Mechanics:** Extended Newton's laws to rotating systems and developed the **Euler equations of motion** for rigid bodies.
- **Celestial Mechanics:** Made major strides on the **three-body problem** (Earth, Moon, Sun), helping to create accurate lunar tables for navigation.
- **Structural Engineering:** Developed the **Euler buckling formula** to determine the critical load a column can support before collapsing.
- **Optics & Music:** Advocated for the **wave theory of light** and developed a mathematical theory of harmony and musical acoustics. [Wikipedia +5](#)

Logic

- **Euler Diagrams:** Created visual tools to represent logical relationships and set intersections (the precursor to Venn diagrams). [Quora +1](#)

Cauchy-Euler Equations

To solve second order Cauchy-Euler differential equations

$$a x^2 y'' + b x y' + c y = 0,$$

we assume

$$y = x^m$$

and we get

$$a m(m-1) + b m + c = 0$$

$$a m^2 + (b-a) m + c = 0$$

this is called **characteristic equation**

Second Order Cauchy-Euler Differential equation

There are three cases for the roots of the characteristic equation

$$a m^2 + (b - a) m + c = 0$$

- ① Two distinct real roots m_1, m_2 , then the general solution is

$$y = c_1 x^{m_1} + c_2 x^{m_2}$$

- ② Repeated real root m_1 , then the general solution is

$$y = c_1 x^{m_1} + c_2 x^{m_1} \ln x$$

- ③ Conjugate complex roots $m_1 = \alpha + i\beta$, $m_2 = \alpha - i\beta$, then the general solution is

$$y = c_1 x^\alpha \cos(\beta \ln x) + c_2 x^\alpha \sin(\beta \ln x)$$

Example

Solve the differential equation

$$2x^2 y'' - 3xy' - 3y = 0, \quad x > 0$$

Solution

$$y = c_1 x^2 + c_2 x^{-\frac{1}{2}}$$

Example

Solve the differential equation

$$x^2 y'' - 3xy' + 13y = 0, \quad x > 0$$

Solution

$$y = x^2 [c_1 \cos(3 \ln x) + c_2 \sin(3 \ln x)]$$

Example

Solve the differential equation

$$x^3 y''' - x^2 y'' - 2xy' - 4y = 0, \quad x > 0$$

Solution

$$y = c_1 x^4 + c_2 \cos(\ln x) + c_3 \sin(\ln x)$$

✓ Example 2.5.4

Find the solution of $x^2 y'' - xy' - 3y = 2x^2$.

Solution

First we find the solution of the homogeneous equation. The characteristic equation is $r^2 - 2r - 3 = 0$. So, the roots are $r = -1, 3$ and the solution is $y_h(x) = c_1 x^{-1} + c_2 x^3$.

We next need a particular solution. Let's guess $y_p(x) = Ax^2$. Inserting the guess into the nonhomogeneous differential equation, we have

$$\begin{aligned} 2x^2 &= x^2 y'' - xy' - 3y = 2x^2 \\ &= 2Ax^2 - 2Ax^2 - 3Ax^2 \\ &= -3Ax^2 \end{aligned} \tag{2.5.5}$$

So, $A = -2/3$. Therefore, the general solution of the problem is

$$y(x) = c_1 x^{-1} + c_2 x^3 - \frac{2}{3} x^2$$

Simultaneous Linear Differential Equations

SIMULTANEOUS DIFFERENTIAL EQUATIONS

If two or more dependent variables are functions of a single independent variable, the equations involving their derivatives are called simultaneous equations, e.g.,

$$\frac{dx}{dt} + 4y = t$$

$$\frac{dy}{dt} + 2x = e^t$$

The method of solving these equations is based on the process of elimination, as we solve algebraic simultaneous equations.

Q.1 The equations of motions of a particle are given by

$$\frac{dx}{dt} + w y = 0$$

$$\frac{dx}{dt} - w x = 0$$

Find the path of the particle and show that it is a circle.

Sol. On putting $\frac{d}{dt} \equiv D$ in the equations, we have

$$Dx + wy = 0 \quad \dots(1)$$

$$-wx + Dy = 0 \quad \dots(2)$$

On multiplying (1) by w and (2) by D , we get

$$wDx + w^2y = 0 \quad \dots(3)$$

$$-wDx + D^2y = 0 \quad \dots(4)$$

On adding (3) and (4), we obtain

$$w^2 y + D^2 y = 0 \quad \Rightarrow \quad (D^2 + w^2)y = 0 \quad \dots(5)$$

Now, we have to solve (5) to get the value of y .

$$\text{A.E. is } D^2 + w^2 = 0 \quad \Rightarrow \quad D^2 = -w^2 \quad \Rightarrow \quad D = \pm iw$$

$$y = A \cos wt + B \sin wt \quad \dots(6)$$

$$\Rightarrow \quad Dy = -A\omega \sin \omega t + B\omega \cos \omega t$$

On putting the value of Dy in (2), we get

$$-wx - A\omega \sin \omega t + B\omega \cos \omega t = 0$$

$$\Rightarrow \quad wx = -A\omega \sin \omega t + B\omega \cos \omega t$$

$$\Rightarrow \quad x = -A \sin \omega t + B \cos \omega t$$

On squaring (6) and (7) and adding, we get

$$x^2 + y^2 = A^2 (\cos^2 \omega t + \sin^2 \omega t) + B^2 (\cos^2 \omega t + \sin^2 \omega t)$$

$$\Rightarrow \quad x^2 + y^2 = A^2 + B^2$$

This is the equation of circle.

Q.2 Solve the following

$$\frac{dx}{dt} = 3x + 8y$$

$$\frac{dy}{dt} = -x - 3y$$

With $x(0) = 6$ and $y(0) = -2$.

Sol. Here we have

$$\frac{dx}{dt} = 3x + 8y$$

$$\frac{dy}{dt} = -x - 3y$$

On putting $\frac{d}{dt} \equiv D$ in (1) and (2), we get

$$Dx - 3x - 8y = 0 \Rightarrow (D - 3)x - 8y = 0 \quad \dots(3)$$

$$\text{and } Dy + x + 3y = 0 \Rightarrow (D + 3)y + x = 0 \quad \dots(4)$$

Multiplying (3) by $(D + 3)$ and (4) by 8 adding them we get

$$(D^2 - 1)x = 0$$

$$\text{A.E. is } m^2 - 1 = 0 \Rightarrow m^2 - 1 \Rightarrow m \pm 1$$

$$\therefore C.F. = C_1 e^t + C_2 e^{-t}$$

$$P.I. = 0$$

$$\therefore x = C_1 e^t + C_2 e^{-t} \quad \dots(5)$$

From (3) we get $(D-3)[C_1e^x + C_2e^{-x}] = 8y$

$$\Rightarrow 8y = -C_1e^t - C_2e^{-t} - 3C_1e^t - 3C_2e^{-t}$$

$$\Rightarrow 8y = -2C_1e^t - 4C_2e^{-t}$$

$$\Rightarrow y = -\frac{1}{2}(C_1e^t + 2C_2e^{-t})$$

Initially when $t = 0$ then $x = 2$.

$$\text{From (5); } 2 = C_1e^0 + C_2e^0 \Rightarrow C_1 + C_2 = 2$$

Also when $t = 0$, $y = -2$.

$$\text{From (6), } -2 = -\frac{1}{4}(C_1e^0 + 2C_2e^0)$$

$$\Rightarrow 8 = C_1 + 2C_2$$

Solving (7) and (8), we get

$$C_1 = -4 \text{ and } C_2 = 6$$

Hence, the required solution is

$$x = -4e^t + 6e^t$$

$$\text{and } y = -\frac{1}{4}(-4e^t + 12e^t)$$

Q.3 Solve : $\frac{d^2x}{dt^2} + y = \sin t$, $\frac{d^2y}{dt^2} + x = \cos t$,

Sol. Here, we have

$$D^2x + y = \sin t \quad \dots(1)$$

$$x + D^2y = \cos t \quad \dots(2) \quad \left(D \equiv \frac{d}{dt} \right)$$

Operating equation (1) by D^2 , we get

$$D^4x + D^2y = -\sin t \quad \dots(3)$$

Subtracting (2) from (3), we get

$$(D^4 - 1)x = -\sin t - \cos t$$

Auxiliary equation is

$$m^4 - 1 = 0 \quad \Rightarrow \quad (m^2 - 1)(m^2 + 1) = 0$$

$$\Rightarrow \quad m = 1, -1, \pm i$$